Journal of Manufacturing Processes 165 (2026) 1–14

Contents lists available at ScienceDirect

Journal of Manufacturing Processes

journal homepage: www.elsevier.com/locate/manpro

Laser powder bed fusion of amitriptyline tablets via eutectic mixtures using sugar alcohols as binders

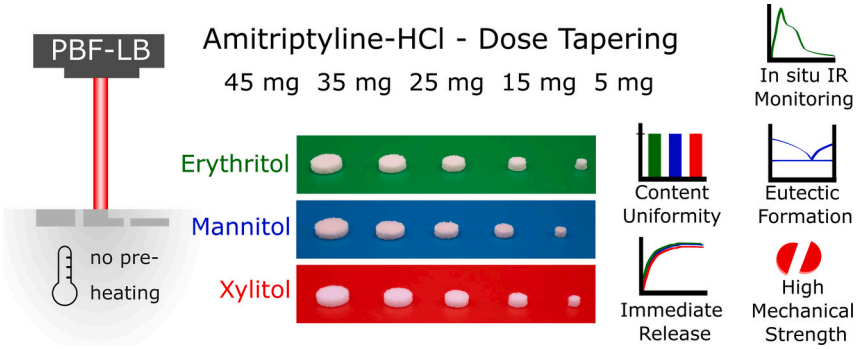
Wessel Kooijman [a,1], Florian Engelsing [a,b,1], Robbert Jan Kok [a], Julian Quodbach [a,*]

[a] Division of Pharmaceutics, Utrecht Institute for Pharmaceutical Sciences (UIPS), Utrecht University, Universiteitsweg 99, 3584, CG, Utrecht, the Netherlands
[b] Glatt Pharmaceutical Services GmbH & Co. KG, Werner-Glatt-Strasse 1, 79589, Binzen, Germany

HIGHLIGHTS

- Sugar alcohols are strong and versatile binders in laser powder bed fusion
- Utilizing eutectic interactions can reduce required laser energy densities
- Tablets are printed without preheating, with the laser as the only energy source
- A robust calibration has been used to print clinically relevant doses of amitriptyline hydrochloride accurately

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Personalized medicine
Laser powder bed fusion
Sugar alcohols
Amitriptyline HCl
Selective laser sintering
Eutectic mixtures

ABSTRACT

Laser powder bed fusion is an additive manufacturing technique with a big potential for producing personalized pharmaceutical dosage forms on an industrial scale. To date, primarily thermoplastic polymers have been used as binding excipients in investigations. This work demonstrates that sugar alcohols erythritol, mannitol, and xylitol are excellent alternatives as binding excipients for laser powder bed fusion in terms of processability, content uniformity, and tablet strength. A robust dose calibration strategy demonstrates that tablets can be compounded with precise and accurate mass by powder bed fusion utilizing a laser as the only source of energy. Manufacturing without preheating reduces energy costs and reduces the risks of thermal degradation of API and excipients, which potentially improves the recyclability of residual powder. Amitriptyline hydrochloride was used as the model drug, as personalized medication of this compound is valuable for therapeutic dosing regimens in the treatment of depressive disorders. Phase diagrams showed that amitriptyline forms eutectic mixtures with sugar alcohols. We show that the formation of eutectic mixtures affords reduction of the laser energy density, thus further lowering the risk of thermal degradation. We demonstrate the great potential of sugar alcohols as binders in laser powder bed fusion with exploitable interactions with amitriptyline hydrochloride.

* Corresponding author.
E-mail address: j.h.j.quodbach@uu.nl (J. Quodbach).
[1] Authors contributed equally.

https://doi.org/10.1016/j.jmapro.2026.02.076
Received 3 December 2025; Received in revised form 17 February 2026; Accepted 26 February 2026
Available online 4 March 2026
1526-6125/© 2026 The Author(s). Published by Elsevier Ltd on behalf of The Society of Manufacturing Engineers. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

1. Introduction

A wide variety of genetic, biological, and lifestyle factors influence how medicines are absorbed, distributed, metabolized, and excreted by the body, affecting their safety and efficacy [1–7]. The personalized medicine concept states that medical therapies should be tailored to patient characteristics, hence affording the proper dosing regimen tailored to optimal pharmacotherapeutic outcome. Manufacturing personalized medicine to supply a larger part of society would require a paradigm shift in industrial manufacturing of oral solid-dose medicines from producing large amounts of "one size fits all" to a variety of different dose strengths. Tablet compression is an optimized process for large-scale tablet production, but it lacks the flexibility needed for compounding such personalized dosage forms. Pharmacies can personalize medication through means of capsule filling or other compounding technologies. However, while effective, these methods are labor-intensive and not scalable for widespread personalization. Recently, additive manufacturing has emerged as a technique for the production of personalized dosage forms [8–11]. Powder bed fusion – laser beam (PBF-LB) has shown particular promise for the scalable production of personalized dosage forms, offering high precision, design flexibility, and suitability for industrial implementation. In the pharmaceutical field, this technique is also known as selective laser sintering (SLS) or laser powder bed fusion (LPBF).

PBF-LB utilizes a laser to fuse powder particles according to a computer model. It operates in a layer-by-layer fashion and is typically used in industry for prototyping or small-scale production of complex parts. A PBF-LB formulation requires two basic components: the active pharmaceutical ingredient (API) and a binder. A variety of polymers, such as copovidone, polyvinyl alcohol, and polyethylene glycol, commonly used as coatings, dissolution modifiers, stabilizers, or hot-melt extrusion binders, have also shown potential as PBF-LB binders [12,13]. They are thermoplastic, cost-effective, possess suitable glass transition temperatures and melting points, and do not undergo chemical changes during processing, making them suitable as binders in PBF-LB. Despite the advantages of these polymers, commonly used thermoplastic polymers have high melt viscosities, limiting coalescence of the material resulting in brittle and weakly bound tablets [14–16]. Additionally, thermoplastic polymers are often processed at elevated temperatures, which alters the material properties leading to reduced recyclability [17]. Expanding the excipient toolbox to include materials with different physicochemical properties would give more options and flexibility in formulation development. Sugar alcohols are promising candidates due to their safety, favorable thermal properties, and pharmaceutical relevance and availability.

Sugar alcohols are derived from monosaccharides or disaccharides through the reduction of carbonyl group(s). Sugar alcohols exhibit a broad melting temperature range (~95–225 °C) and are well suited for PBF-LB [18]. Their melt viscosity (10^{-2} -10^{0} Pa·s) is significantly lower than that of commonly used thermoplastic polymers (10^{3}–10^{6} Pa·s), suggesting that the molten sugar alcohols have improved coalescence and material flow to densify and bind powder particles during PBF-LB [14–16,19]. Mannitol's potential as a binder in PBF-LB has been demonstrated by Gotoh et al., while Allahham et al. and Stanojevic et al. have investigated its use as a filler [20–22]. Horváth et al. showed that sorbitol can function as a melt binder in twin-screw melt granulation [23]. Sugar alcohols are also commonly used as binders/fillers in tablet compression [24,25].

Sugar alcohols are commonly used as food additives and can be found in various products such as cereal bars, toothpaste, and baby formula. Naturally, they also occur in many foods, for example mushrooms, cauliflower, and cheese. Sugar alcohols provide a similar level of sweetness to sugar but with a reduced caloric value. Their safety was established in 1994 by the European Parliament and Council, which has assigned them an acceptable daily intake of "quantum satis", which means that usual levels of food additives do not represent a health concern [26]. Erythritol has recently been reevaluated by the European Food Safety Authority, and an acceptable daily intake of 0.5 g/kg was established to prevent immediate laxative effects and potential chronic effects [27]. Mannitol and xylitol are part of the same series of evaluations, but these evaluations are still ongoing as of the writing of this publication [28]. Excessive intake of mannitol and xylitol (> 20 g per day) can cause gastrointestinal discomfort, most notably laxation [29,30]. The established daily intake limits are well above the quantities that would be present in tablets, confirming their safe applicability in this context. More recently, *Hazen* et al. reported a correlation between xylitol and erythritol intake and an increased risk of major adverse cardiovascular events [31,32]. No direct causation was shown but new findings on the long-term safety of xylitol and erythritol should be taken into account in regard to this work.

The safety and versatile applicability of sugar alcohols leads to a high demand in the food and pharmaceutical industries, resulting in a wide range of chemical and physical grades available on the market. This diversity is advantageous for PBF-LB, which requires binders with specific melting temperatures and particle sizes. Additionally, their varying levels of sweetness can serve as effective taste masking agents. Sugar alcohols are also highly water-soluble, making them ideal for immediate-release dosage forms. To investigate the practical application of sugar alcohols as binders in PBF-LB, this study investigates their use in formulations for the delivery of the API amitriptyline hydrochloride (AMT).

AMT is a tricyclic antidepressant used in psychiatric disorders. It works through inhibition of serotonin and norepinephrine transporters, thus prolonging the activation of serotonergic and adrenergic neurons. There are numerous indications for which AMT is prescribed, such as depression, chronic pain management, including obsessive–compulsive disorder, panic attacks, generalized anxiety disorder, post-traumatic stress disorder, bulimia nervosa, smoking cessation, or enuresis [33]. Depending on the indication and patient, tablet dosages can vary between 10 and 150 mg for patients [34]. The available tablet dosages are 10, 25, 50, 75, and 100 mg [35].

Due to the risk of withdrawal, antidepressants like AMT often require careful tapering when medication is changed or discontinued. Changing the drug dose in a hyperbolic tapering method, with daily dose reductions averaging around 4.5% (a few mg), has been shown to significantly reduce withdrawal symptoms [36,37]. The required variety in different doses is challenging to produce with conventional manufacturing and additive manufacturing would be an ideal manufacturing method to produce them. Additionally, AMT is considered to be a narrow therapeutic index drug, meaning that its dosing and the resulting drug levels in each patient need to be guided by therapeutic drug monitoring (TDM), in which individual blood concentrations are leading in the personalized dosing regimen in order to avoid therapeutic failure or adverse drug reactions [38]. The small therapeutic window and need for patient-specific dosages underline the potential for additive manufacturing to prepare these personalized dosages of AMT and other tricyclic antidepressants like nortriptyline and doxepin.

The goal of this study is to investigate the capability of sugar alcohols to act as binding excipients in PBF-LB without preheating. AMT was formulated with erythritol, mannitol, and xylitol in three formulations, which were then manufactured in five clinically relevant doses (5, 15, 25, 35, 45 mg). The dosage forms were evaluated using the following pharmacopeial tests: content uniformity, friability, tensile strength, disintegration, and dissolution testing. Furthermore, the influence of the thermal and solid-state properties of the sugar alcohols on their behavior during PBF-LB and the structure of the final dosage forms was investigated. DSC, TGA, XRPD, and IR imaging were utilized to characterize these properties and their relationships.

2. Materials and methods

2.1. Materials

D-mannitol (Pearlitol® 100SD Mannitol, Roquette, France), xylitol (Xylisorb® 90 Xylitol, Roquette), and erythritol (non-disclosed) were used as binders. AMT (Amitriptyline HCl, Duchefa, the Netherlands) was used as a model API, while colloidal silica (Aerosil 200, Evonik, Germany) was added as a glidant. The ratio of the components can be found in Table 1. All materials were sieved through a 200 μm sieve to remove agglomerates and large particles. The mannitol formulation was prepared by mixing mannitol and AMT for 10 min at 72 rpm, after which the blend was sieved through a 315 μm mesh. The xylitol and erythritol formulations were prepared by mixing the erythritol or xylitol with AMT for 10 min at 72 rpm, after which the colloidal silica was mixed in for 10 min at 49 rpm. The resulting blend was sieved over a 315 μm sieve. The particle size distribution of the final blends can be found in **Fig. S1**. To clearly distinguish between the pure substances and the formulations, the mixtures containing erythritol, mannitol, and xylitol with 15 wt% AMT are referred to as ERY-A, MAN-A, and XYL-A, respectively. MAN75-A refers to the formulation used to study the manufacturing behavior of eutectic mixtures.

2.2. Tablet design

Cylindrical tablets were designed in AutoCAD® 2024 (Autodesk, USA). To target the desired dose, the tablet diameters were adjusted according to the calibration curve described in section 3.1.3. The tablet height was kept constant at 3 mm. In Table 2, the dosages and respective dimensions are listed. The tablets were sliced in the EOSprint 2.8 software (EOS group, Germany) and evenly spaced on the build platform with a minimum distance of 2 mm between the tablets.

2.3. Manufacturing and sampling

A modified AMCM M290 (EOS, Germany) with a 5 W CO_2 laser was used. Approximately 500 g of formulation was loaded in the powder reservoir, after which the build chamber was flushed with N_2 gas. Manufacturing was performed without preheating of the powder; the only energy input came from the laser. The build chamber was at ambient temperature and was not specifically controlled. The first layer was directly fused on the metal build plate, resulting in the tablets adhering to the build plate. The tablets were manufactured with a hatch pattern and hatch spacing of 100 μm. Contour settings were deliberately omitted to avoid edge effects. The layer height was set to 200 μm. The scanning speed and laser power were varied for each formulation. From these parameters, it is possible to calculate the energy density (ED) using Eq. 1,

$$ED = \frac{P}{vht} \quad (1)$$

where P is the laser power (W), v is the scanning speed of the laser (mm/s), h is the hatch spacing (mm), and t is the layer thickness (mm). The laser power, scanning speed, and energy densities used for the different formulations are displayed in Table 3.

After manufacturing, XYL-A tablets required 10 min to completely solidify before separation from the build plate and cleaning with a paintbrush. MAN-A and ERY-A tablets solidified immediately and were separated from the build plate directly, without waiting time at the end of the process. The remaining powder in the powder reservoir and overflow container was collected and stored for further analysis.

Table 1
Weight ratios of AMT and excipients in the formulations.

Formulation	Erythritol (ERY-A)	Mannitol (MAN-A)	Xylitol (XYL-A)	Mannitol (MAN-75 A)
Sugar alcohol	84%	85%	84%	24.75%
Amitriptyline HCl	15%	15%	15%	74.25%
Colloidal silica	1%	–	1%	1%

Table 2
Dimensions of the manufactured cylindrical tablets with varying diameters (mm) and constant height of 3 mm for each formulation.

Desired dose (mg)	Tablet diameter (mm)		
	ERY-A	**MAN-A**	**XYL-A**
5	4.7	4.4	4.6
15	7.0	6.9	7.0
25	8.8	8.6	8.7
35	10.2	10.1	10.1
45	11.5	11.4	11.4

Table 3
Process parameters for the formulations.

	ERY-A	MAN-A & MAN-75 A	XYL-A*	MAN-A & MAN-75 A
Scanning speed (mm/s)	375	250	450	100
Laser power (W)	5.0	5.0	4.0	1.0
Energy density (J/mm^3)	0.67	1.00	0.44	0.5

* The first two layers are manufactured at two times the scanning speed.

2.4. Characterization

2.4.1. Particle size analysis

The particle size distributions were determined using a Mastersizer 3000 (Malvern Panalytical, United Kingdom, and the Netherlands) equipped with an Aero S dry powder dispersion unit. The air pressure was 1 barg for all the materials measured except for the measurement of the pure AMT, where the air pressure was set to 1.5 barg. The feed rate was set at 40% for all materials, while the hopper gap was varied between 0.5 and 1.5 mm to achieve an obscuration rate between 0.5% and 8%. The refractive index was set to 1.5, and measurements were performed in triplicate.

2.4.2. Solid state analysis

Prior to solid state analysis, the produced tablets were manually crushed in a mortar and pestle for several minutes.

Degradation temperatures were determined with a TGA Q50 (TA Instruments, USA). Samples of 10 to 15 mg were weighed and placed in a platinum pan. The samples were heated from 40 °C to 400 °C at a heating rate of 5 °C/min under a nitrogen flow of 60 ml/min. The degradation temperature was defined as the temperature at which the weight loss rate exceeded 0.1% of the original mass per minute.

To assess thermal properties of the samples, a Discovery DSC (TA Instruments, USA) was used. Samples of 5–15 mg were weighed and placed in 20 μl aluminum pans and sealed with non-hermetic lids. The samples were heated at a heating rate of 5 °C/min from 20 to 80 °C for an initial heating cycle to remove any moisture after which they were cooled to 0 °C at 5 °C/min. In a second heating cycle the samples were heated to 200 °C at a heating rate of 5 °C/min. When analyzing AMT, the temperature range was extended to 225 °C to capture the full melting peak. The measurements were performed under a nitrogen flow of 50.0 ml/min. The crystallization and melting points are reported as onset temperatures. Phase transition enthalpies and onset temperatures are reported as the mean ± SD ($n = 3$).

To construct binary phase diagrams, the Discovery DSC was used to study the melting behavior of mixtures of sugar alcohols with AMT.

Mixtures of the sugar alcohols with AMT were prepared in 10 wt% increments, and additional ratios were included to improve the resolution and accuracy of the phase diagram fitting closer to the eutectic point. DSC measurements were performed as described above. The solidus temperature was defined as the onset temperature of the first melting peak (appearance of the first liquid). The liquidus was defined as the peak temperature of the second peak (disappearance of the last solid). To construct the solidus lines, the average of all experimentally determined solidus temperatures was used. The experimentally determined liquidus temperatures were fitted with the Schröder-van-Laar (SVL) equation:

$$ln(x_1\gamma_1) = -\left(\frac{\Delta H_f}{R}\right) * \left(\frac{1}{T} - \frac{1}{T_m}\right) \quad (2)$$

where x_1 is the mole fraction of component 1, γ_1 is the activity coefficient, $\Delta H_{f,1}$ is the molar enthalpy of fusion, $T_{m,1}$ is the melting temperature of component 1, T is the liquidus temperature, and R is the gas constant [39,40]. This equation can be applied symmetrically to both components in a binary system. For each side of the eutectic, the equation models the solubility of one component in the melt of the other. By switching the roles of components 1 and 2 (using $x_2 = 1 - x_1$, γ_2, $\Delta H_{f,2}$, $T_{m,2}$), the liquidus curve for the second component can be calculated. γ_1 and γ_2 are defined as shown in Eqs. 3 and 4, yielding four fitting parameters:

$$\ln(\gamma_1) = ax_1^2 + bx_1 \quad (3)$$

$$\ln(\gamma_2) = c(1 - x_1)^2 + d(1 - x_1) \quad (4)$$

Both liquidus curves were fitted to pass through a common intersection with the experimentally determined solidus, thereby providing a predictive model for the eutectic point. This method of data fitting allows for great flexibility but lacks a theoretical background; the fitting parameters a, b, c, and d therefore have limited physical meaning.

X-ray powder diffraction (XRPD) patterns were recorded using a D2 PHASER (Bruker, USA) with a Lynxeye detector using Co K_α radiation. Measurements were taken with $2\Theta = 10–60°$ at 0.01° increments with 1 s per increment. The poly(methyl methacrylate) sample holder rotated at 15 rpm.

2.4.3. Dosage form analysis

The weight of the tablets was determined using a Secura 224–1 s balance (Sartorius, Germany). The physical dimensions (diameter and height) and hardness were measured using a SOTAX MT50 hardness tester (Sotax, Switzerland). The tensile strength of the tablet was then calculated using Eq. 5:

$$T = \frac{2F}{\pi D t} \quad (5)$$

where F is the diametrical breaking force, D is the tablet diameter, and t is the tablet thickness [41].

The disintegration time of the tablets was determined with a single basket tablet disintegration tester (PTZ-S, Pharmatest, Hainburg, Germany) according to European Pharmacopeia 2.9.1 ($n = 6$).

Drug content was determined by dissolving ten samples per condition in 100 ml distilled water. The samples were diluted, and the drug concentration measured using ultra-performance liquid chromatography (UPLC, Waters Acquity UPLC H-Class System). The system was equipped with an RP 18 column (Acquity HSS T3 1.8 µm 2.1 × 50 mm) and a UV/Vis-detector. A gradient method was applied, using a 4:1 mixture of 25 mM sodium phosphate buffer (pH 3) and acetonitrile as mobile phase A and a 1:1 ratio for mobile phase B. The gradient program started at 0% B, held for 0.5 min, then increased linearly to 100% B over 5 min, maintained for 1.0 min, and returned to initial conditions over 0.1 min, followed by a 2 min equilibration. The column was kept at a temperature of 50 °C, at a flow rate of 1 ml/min. All AMT samples were quantified as the area under the curve (AUC) of the UV absorbance peak at a wavelength of 240 nm, after a 10 µl sample injection. The AMT standard curve was linear in the range of 1 to 60 µg/ml ($R^2 > 0.999$). The acceptance value (AV) was calculated according to Ph. Eur. 2.9.40 Uniformity of dosage units.

A one-way analysis of variance (ANOVA) test was done to compare the means of content of the 25 mg dose tablets using Prism version 10.4.1 (GraphPad Software, USA).

The tablets were subjected to in vitro drug release studies using a dissolution apparatus (PT-PDT7, Pharmatest, Germany). The dissolution vessels were filled with 500 ml of 25 mM potassium phosphate buffer, pH 6.8. The temperature was set to 37 °C and the paddle speed to 75 rpm. A 1 ml sample volume was withdrawn after 2, 5, 10, 15, 30, and 60 min. The dissolution volume was kept constant throughout the experiment by replacing the withdrawn sample with phosphate buffer medium. The studies were performed in triplicate and analyzed using the same UPLC method as specified before.

2.4.4. Imaging

Images of tablets were taken with a Canon EOS R50 camera with a Canon RF-S 18-45 mm F/4.5–6.3 IS STM lens (Canon, Japan) inside a 30.5 × 30.5 × 30.5 cm photography light box (Duclus).

A PI 640i IR camera (Optris, Germany), equipped with a 15° x 11° field of view lens and a spectral filter with a cut off at 11.5 µm to protect the long-wave sensor from CO_2 laser damage, was used to measure the temperature of the powder bed during the process. The detector has a resolution of 640 × 120 pixels, and the videos were recorded at a frequency of 125 Hz. The IR camera was positioned outside of the build chamber at a height of ~450 mm above the powder bed and with an angle of ~12°. A zinc-selenide viewing window (Crystran, United Kingdom) with ~70% transmission across 0.6–14 µm was used. The camera was manually focused by maximizing the edge sharpness of a piece of tape on the build plate. The ambient temperature value was provided by the camera's internal sensor. Emissivity values were determined using a hot stage (FTIR 600, Linkam, United Kingdom) positioned inside the build chamber on top of the build plate (**Fig. S2**). Thin powder layers of each formulation were heated to the melt and subsequently cooled. The emissivity value was adjusted to match the hot stage set temperature across the temperature range, yielding formulation-specific emissivity values ($\varepsilon_{ERY\text{-}A} = 0.63$, $\varepsilon_{MAN\text{-}A} = 0.65$, $\varepsilon_{XYL\text{-}A} = 0.62$, $\varepsilon_{MAN\text{-}75A} = 0.62$). A reference measurement point with a fixed temperature at 23 °C was set on unheated powder at the side of the build platform. Measured values and relative deviations between IR readings and set temperatures are shown in **Table S1**. During manufacturing, the final, uncovered layer was captured for every formulation. Temperature-time profiles were created using PIX connect version 3.25.3131.0 software (Optris, Germany). Twenty pixels distributed over four tablets were used to track the temperature over time during manufacturing, yielding twenty temperature profiles. The temperature profiles were smoothed with a Savitzky–Golay filter (3rd degree polynomial, 21-point window width). Temperature peaks were determined after the smoothing filter was applied, which improved the measurement reproducibility but slightly reduced the found peak temperatures (2–10 °C reduction). To quantify the crystallization enthalpy observed for ERY-A and MAN-A, a temperature decay function (Eq. 6) was fitted to the smoothed temperature profiles, excluding the region where the crystallization peak was observed. The temperature difference between the fitted temperature decay function and smoothed temperature profile was integrated over time to yield the enthalpy of fusion (ΔH) (or crystallization) (Eq. 7).

$$T(t) = \frac{a}{(t+b)^c} + d \quad (6)$$

$$\Delta H = \frac{Q}{m} = c_p \int \left(T - T_{fit}\right) dt \quad (7)$$

where, T and T_{fit} represent the measured and fitted temperatures (°C) respectively, t is time (s), C_p is the specific heat capacity (J/g°C^{-1}), m is the sample mass (kg), Q is the total heat released (J) and ΔH is the effective enthalpy of crystallization.

The temperature deviation attributed to crystallization is mostly due to latent heat release while the sample cools in its solid state. Using the C_p for the solid-state materials best represents the thermal response of the material during and immediately after crystallization. As data on the heat capacity of AMT was unavailable, it was assumed that the sample consisted of the pure sugar alcohols. The C_p of mannitol at 160 °C and erythritol at 100 °C were taken from literature and are 1815 and 1910 J kg^{-1} K$^{-1,}$ respectively [42]. The C_p is temperature dependent but was assumed to be a constant for simplicity. The reported peak temperatures and crystallization enthalpies are averages over twenty temperature profiles.

Scanning electron microscope (SEM) images were made of the exterior and cross-sections of the tablets using a Phenom ProX (Thermo Fischer Scientific, USA). The uncoated samples were placed on a charge reduction sample holder. The samples were measured with an acceleration voltage of 10 kV and a back-scattering detector. ImageJ version 1.54 g (Wayne Rasband, NIH, USA) was used to determine pore diameters from the SEM images.

3. Results and discussion

3.1. Manufacturing of oral dosage forms

3.1.1. Powder characteristics and spreading

For successful PBF-LB manufacturing, a powder formulation requires good spreadability, which is the ability to form a uniform layer [43]. Spreadability is closely connected to powder flowability, and therefore to particle size and shape. The powder spreading behavior was assessed visually during the build process. For MAN-A powder, spreading was sufficient without the addition of colloidal silica due to the nodular shape of the spray-dried mannitol (Fig. 1B). 1% of colloidal silica had to be added to the ERY-A and XYL-A formulations to improve flowability as the irregular shaped ERY and XYL particles (Fig. 1A,C) prevented proper spreading on the build platform.

3.1.2. PBF-LB without preheating

The tablets were manufactured without preheating, with the laser serving as the sole energy source to fuse the powder. The absence of powder bed heating reduces energy costs and thermal stress on the unscanned powder [44]. This prevents unprocessed powder aging and potential degradation, allowing for simplified powder recycling [17]. Minimizing the formation of degradation products is particularly important in this context, as the final product is intended for human consumption. When the laser melts the first powder layer, a large temperature gradient is created, which increases the likelihood of droplet formation. The molten material coalesces into larger spheres instead of wetting the underlying powder layer [45]. When the next layer is spread the melt droplets are dragged by the recoater through the powder bed. To overcome this, the first layer was directly fused on the build platform. Since the melt wets the steel surface better than the powder particles, droplet formation and dragging were eliminated. The laser power (P) and scanning speed (v) were adjusted for each formulation to produce tablets with sufficient mechanical strength. The resulting energy densities increase from XYL-A (0.44 J/mm^3) to ERY-A (0.67 J/mm^3) to MAN-A (1.00 J/mm^3), reflecting the melting points of the respective sugar alcohol binder. For the first time, we demonstrated that API-containing tablets can be produced using PBF-LB without preheating the powder bed.

3.1.3. Volume-based mass calibration

Compared to conventional tablet compaction, additive manufacturing allows for patient-specific dosing by adjusting the tablet geometry without modifying the pharmaceutical formulation. To precisely manufacture five clinically relevant AMT tablet doses, a first set of cylindrical tablets with diameters ranging from 4 to 12 mm and a constant height of 3 mm were produced to generate a calibration curve covering a broad volume range. Assuming a 15% AMT concentration (*w*/w), a second set of tablets targeting doses of 5, 15, 25, 35, and 45 mg was produced. A strong linear correlation between tablet weight and STL volume was observed within this range, as indicated by the R^2 values (> 0.9995) (Fig. 2). Each dose was prepared as a single batch, with the 25 mg tablets manufactured in triplicate to assess inter-batch variability. Unlike earlier approaches that assumed a constant density and a calibration line through the origin, a size-dependent density was observed resulting in calibration lines with a nonzero intercept, thereby justifying this approach [46,47].

3.2. Tablet characterization

3.2.1. Macroscopic and microscopic structure

The five tablet doses produced from each formulation are shown in Fig. 3 alongside the corresponding SEM images. ERY-A and XYL-A tablets had a uniform white appearance, whereas the MAN-A tablets showed a slight discoloration, which is not visible in the images. It is well known that mannitol turns yellow as a result of thermal degradation [48]. As most organic molecules absorb at the emitted wavelength of a CO_2 laser ($\lambda = 10.6$ μm), no external absorbent was needed. This contrasts with diode laser PBF-LB processes, which are more commonly investigated for pharmaceutical applications, where absorbents are necessary [17,44,49]. The ERY-A tablets (Fig. 3A) have a dense core,

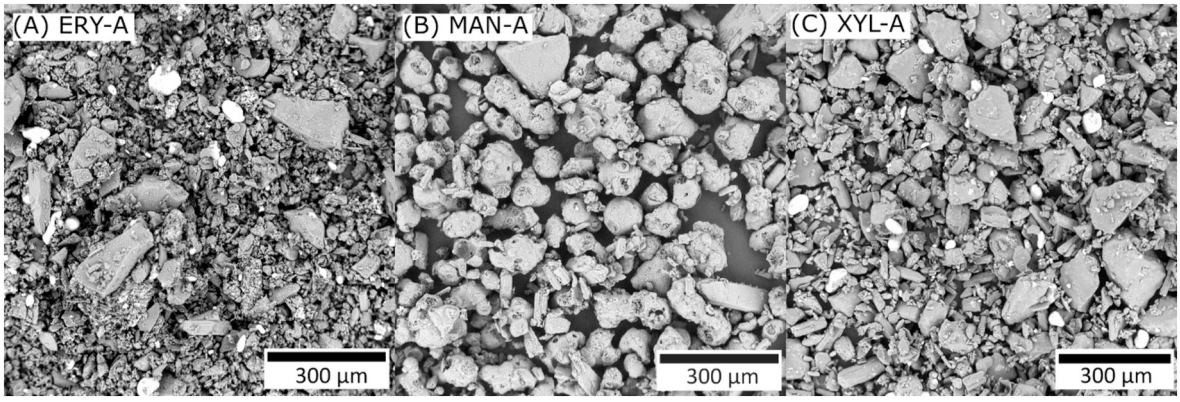


Fig. 1. SEM images of ERY-A (A), MAN-A (B), and XYL-A (C) formulations and corresponding D10, D50, D90 and span values determined by laser diffraction analysis. (**Fig. S1**).

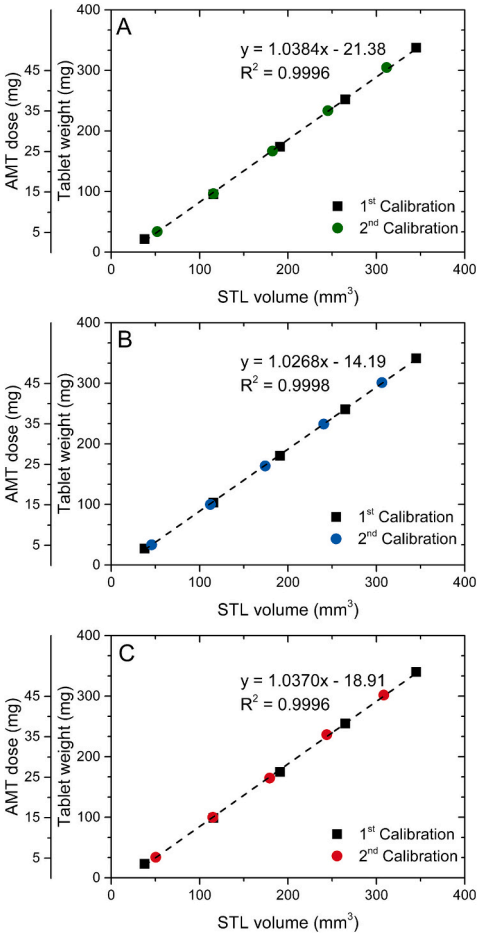


Fig. 2. Calibration line for ERY-A (A), MAN-A (B), and XYL-A (C) correlating tablet STL volume with tablet weight. For the first calibration set, 3D tablet models spanning a wide STL volume range were designed, manufactured via PBF-LB, and the resulting tablets were weighed. The second y-axis shows the theoretical AMT dose, assuming an AMT content of 15% (*w*/w) of the tablet weight. For the second calibration set, patient-relevant doses (5, 15, 25, 35 and 45 mg) were targeted. Both tablet sets were then combined to generate a final calibration line, which was used to produce dose-specific tablets in batches.

whereas the surface is rough with particles loosely fused to the surface. Similarly, the MAN-A tablets (Fig. 3B) have a rough surface and a dense tablet core with a needle-like crystallization patterns**,** which is more clearly visible in the zoomed-in scan in **Fig. S3**. Both ERY-A and MAN-A tablets have small pores which are visible as black voids with widths of 10 to 50 µm likely due to gas entrapment [50]. In contrast, the XYL-A tablets (Fig. 3C) have a porous structure, characterized by larger channel-like voids measuring up to 1 mm in diameter.

3.2.2. Mass and content uniformity

Mass and content uniformity are shown in Table 4. The mass uniformity was excellent across the majority of doses, with relative standard deviations (RSD) mostly below 1%. However, the three 5 mg tablet batches, i.e. the smallest tablets, showed slightly higher deviation with RSDs of 1.6% (ERY-A), 1.3% (MAN-A), and 2.3% (XYL-A). As noticed in section 3.2.1, particles that stick out from the tablet sides can detach during handling, causing variations in tablet weight. This affects lighter tablets more than heavier ones. Despite this, the weight deviations are low, demonstrating that both PBF-LB and sugar alcohol formulations are suitable for producing tablet batches with precise weight control across a range of dose strengths.

The content uniformity was assessed according to Ph. Eur. 2.9.40, with the aim of obtaining acceptance values $\leq$15.0 to demonstrate acceptable content uniformity. All tablet batches met this criterion except the XYL-A 5 mg tablets, which deviated by 7.51% from the target drug load and showed a high RSD, resulting in non-conformity. Across all dose strengths, XYL-A tablets revealed the highest content variability with RSD values ranging from 2.68% to 5.59%. Given the low weight variation, this variability likely resulted from an inhomogeneous powder blend. To verify this hypothesis, ten powder samples taken from different positions within the storage container were analyzed by UPLC, revealing a content of 98.07 $\pm$ 3.07%, supporting the hypothesis of poor blend homogeneity. In contrast, the ERY-A and MAN-A powder samples had lower content deviation. Correspondingly, the ERY-A and MAN-A tablet batches showed good accuracy with low deviation from the targeted dose.

A one-way ANOVA was conducted to assess the inter-batch variability among the three 25 mg tablet batches for each formulation. For the MAN-A and XYL-A tablet batches, no significant differences were found (MAN-A: F = 1.127, p = 0.3387; XYL-A: F = 2.315, p = 0.1181). For the ERY-A formulation, the comparison of the 25 mg tablet batches revealed that batch B1 had a significantly higher mean content (25.9 $\pm$ 0.4 mg) compared to batch B3 (25.3 $\pm$ 0.7 mg) (ERY-A: F = 3.436, p = 0.0468). However, as all ERY-A 25 mg had AV values far below 15, this significant difference is not relevant in practice.

3.2.3. Physical and mechanical characterization of tablets

Tablets require sufficient mechanical strength to withstand coating, packaging, and transport stresses, typically targeting tensile strength $\geq$1.7 MPa and friability <1% [51]. The tablets manufactured via PBF-LB using sugar alcohol binders achieved high tensile strength (ERY-A $\geq$ 2.61 MPa; MAN-A $\geq$ 4.33 MPa; XYL-A $\geq$ 1.48 MPa) and low friability (ERY-A = 0.92%; MAN-A = 0.62%; XYL-A = 1.86%) while maintaining rapid disintegration within 10 min (Table 5). Consequently, ERY-A and MAN-A tablets meet conventional requirements for compacted tablets. Out of the three formulations, the MAN-A tablets were the strongest and took the longest to disintegrate, followed in order by ERY-A and XYL-A. The MAN-A and ERY-A tablets eroded from the outside, while the XYL-A tablets fully disintegrated once water penetrated the large channel-like voids in the tablet (see Fig. 3C). The smaller tablets have a larger surface-to-volume ratio, so they disintegrated faster. Interestingly, the tablets with the smallest diameter, targeting a dose of 5 mg, exhibited high tensile strength. As previously discussed by Lura et al., the hardness tester might not be suitable to accurately measure the breaking force of minitablets, as they do not always break with a diametral crack but rather get squeezed between the metal plates [52].

3.2.4. Drug release

Only the 25 mg dose of each formulation was tested, and the results are shown in Fig. 4. More than 80% of the AMT was released within 10 min for all three formulations, demonstrating immediate release characteristics. The fast dissolution profile can be explained by the high solubility of the AMT hydrochloride salt and of the sugar alcohols. The latter typically show good water solubility, due to the hydrogen bonding between the hydroxyl groups of the sugar alcohols and water. Additionally, solutions of sugar alcohols have a low viscosity, not impeding dissolution [53]. While we only tested the 25 mg dosage forms, it is expected that the dissolution profiles of the other doses will also display immediate release characteristics for the same reasons outlined above.

Comparing the three sugar alcohol formulations, XYL-A tablets dissolved the fastest, followed by ERY-A and then MAN-A. The differences in dissolution rate correspond to their respective solubility (0.63 g XYL, 0.33 g ERY, 0.18 g MAN in 1 g H_2O, respectively) and disintegration time (Table 5) [54]. The findings are consistent with Narala et al. and Pavar et al., who investigated sugar alcohol-based solid dispersions manufactured via hot-melt extrusion and similarly observed that XYL-A tablets dissolved faster than MAN-A tablets [55,56]. Our dissolution data demonstrate sugar alcohols, due to their hydrophilic nature, are suitable excipients for immediate-release dosage forms in

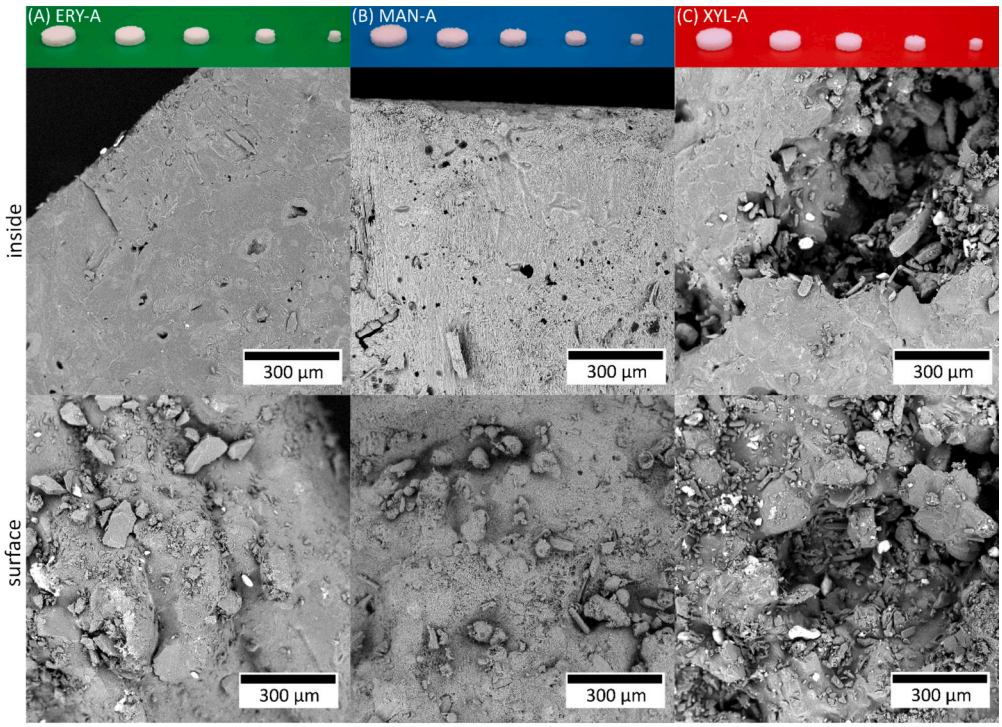


Fig. 3. The five manufactured tablet sizes and SEM images of ERY-A (A) MAN-A (B) and XYL-A (C) tablets.

Table 4
Tablet mass ($32 \leq n \leq 118$) reported as mean, standard deviation (SD), and relative standard deviation (RSD), and AMT content ($n = 10$) reported as percentage of the declared dose, standard deviation, and corresponding acceptance value (AV).

	Declared dose	Tablet mass ± SD [mg]	Tablet mass RSD [%]	AMT content ± RSD [%]	Acceptance value; pass if ≤15
ERY-A	**5 mg**	34.2 ± 0.6	1.6	101.28 ± 2.49	6.1
	15 mg	97.0 ± 0.9	1	98.83 ± 1.69	4.0
	25 mg B1	168.5 ± 1.4	0.8	102.73 ± 1.49	4.9
	25 mg B2	167.6 ± 1.4	0.9	101.18 ± 1.64	4.0
	25 mg B3	168.2 ± 1.5	0.9	100.35 ± 2.75	6.6
	35 mg	235.3 ± 1.9	0.8	101.37 ± 1.26	3.1
	45 mg	306.2 ± 2.9	0.9	102.78 ± 1.46	4.9
	Powder	–	–	102.78 ± 1.92	–
MAN-A	**5 mg**	34.1 ± 0.4	1.3	101.96 ± 1.59	4.3
	15 mg	100.7 ± 0.8	0.8	101.31 ± 1.84	4.5
	25 mg B1	164.3 ± 1.3	0.8	98.54 ± 1.43	3.4
	25 mg B2	165.2 ± 1.3	0.8	98.60 ± 1.14	2.7
	25 mg B3	165.4 ± 1.3	0.8	97.53 ± 2.60	7.1
	35 mg	243.3 ± 2.0	0.9	99.04 ± 1.80	3.7
	45 mg	303.3 ± 3.2	1.1	99.70 ± 2.00	4.8
	Powder	–	–	99.22 ± 1.26	–
XYL-A	**5 mg**	34.3 ± 0.8	2.3	107.51 ± 5.59	20.4
	15 mg	99.3 ± 1.4	1.4	101.87 ± 4.43	11.2
	25 mg B1	165.9 ± 1.4	0.8	101.03 ± 5.10	12.4
	25 mg B2	165.4 ± 1.7	1.0	97.63 ± 2.68	7.1
	25 mg B3	166.2 ± 1.4	0.9	101.37 ± 4.64	11.3
	35 mg	239.4 ± 1.7	0.7	102.88 ± 4.77	13.2
	45 mg	305.6 ± 2.4	0.8	100.48 ± 4.86	11.7
	Powder	–	–	98.07 ± 3.07	

pharmaceutical PBF-LB manufacturing.

3.3. Solid-state characterization

In addition to the pharmaceutical evaluation, the manufacturing process and resulting tablets were examined from a physicochemical perspective. First, the supercooling and crystallization behavior of the formulations was characterized. Second, the potential for thermal degradation was assessed, and, finally, the eutectic interactions between AMT and the sugar alcohols were investigated using DSC.

Table 5
Tablet volume (mean ± SD; $n = 10$), tensile strength (mean ± SD; n = 10), friability (n = 1), and disintegration time (mean ± SD; $n = 6$) of the ERY-A, MAN-A and XYL-A tablets manufactured at five doses.

	Target dose	Volume [mm^3] (n = 10)	Tensile strength [MPa] (n = 10)	Friability [%] (n = 1)	Disintegration time [min] (n = 6)
ERY-A	5 mg	34.1 ± 3.0	3.70 ± 1.25	–	1.31 ± 0.10
	15 mg	87.4 ± 3.9	2.61 ± 0.31	–	2.63 ± 0.07
	25 mg	149.4 ± 7.2	2.65 ± 0.35	0.92	3.28 ± 0.10
	35 mg	204.6 ± 6.6	3.12 ± 0.82	–	3.37 ± 0.06
	45 mg	265.0 ± 5.0	3.33 ± 0.91	–	3.50 ± 0.06
MAN-A	5 mg	32.2 ± 1.1	5.70 ± 1.08	–	2.26 ± 0.49
	15 mg	96.3 ± 3.7	4.48 ± 0.89	–	4.06 ± 0.07
	25 mg	156.3 ± 4.9	4.33 ± 0.52	0.62	4.59 ± 0.12
	35 mg	222.0 ± 5.5	4.57 ± 0.78	–	5.15 ± 0.07
	45 mg	286.4 ± 6.4	4.69 ± 1.11	–	5.51 ± 0.08
XYL-A	5 mg	33.9 ± 2.0	2.02 ± 0.58	–	1.17 ± 0.08
	15 mg	93.9 ± 3.3	1.70 ± 0.32	–	1.36 ± 0.06
	25 mg	152.2 ± 2.7	1.61 ± 0.22	1.86	1.65 ± 0.18
	35 mg	221.2 ± 5.8	1.49 ± 0.26	–	1.74 ± 0.15
	45 mg	280.4 ± 6.0	1.48 ± 0.11	–	1.78 ± 0.09

3.3.1. Sugar alcohol supercooling and crystallization behavior

Sugar alcohols are well known for their supercooling behavior, which is studied extensively in the context of phase change materials for potential use in heat batteries [18,57–59]. Fig. 5 shows the DSC thermograms of the heating and cooling cycles of ERY-A, MAN-A, and XYL-A. ERY-A and XYL-A remain amorphous upon cooling, indicated by the absence of an exothermic peak, while MAN-A crystallizes at a significantly lower temperature (121.9 ± 0.4 °C) than the melting temperature of its main melting peak (161.7 ± 0.3 °C). The IR camera thermograms

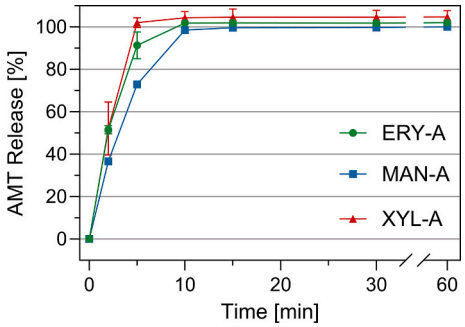


Fig. 4. Release profiles of the ERY-A (circle), MAN-A (square), and XYL-A (triangle) tablets, in phosphate buffer (pH 6.8) at sink-conditions for 0–60 min. Mean ± SD (n = 3).

revealed that both ERY-A and MAN-A crystallized upon cooling after the laser had passed, at approximately 120 ± 5 °C and 165 ± 16 °C, respectively (Fig. 6A,B). This is evident from the exothermic crystallization peak, which appears as a shoulder on the cooling curve and delays cooling of the material. The crystallization temperatures are close to the melting temperatures of pure erythritol (118.8 ± 0.3 °C) and mannitol (166.1 ± 0.8 °C) in these formulations. The discrepancy in crystallization behavior and temperatures observed between the IR imaging during manufacturing and the DSC measurements can be attributed to differences in both the thermal environment and sample conditions. During the manufacturing process, the molten material is surrounded by unsintered powder and underlying solidified layers, which may act as nucleation sites and promote earlier crystallization. However, the significantly different heating and cooling rates of the laser-based process compared with the controlled thermal profile of the DSC also play an important role. In the DSC, the sample is fully molten and lacks residual solids that could serve as seed crystals, and the slower cooling can allow the melt to remain supercooled for longer. Thus, both the thermal history and the presence (or absence) of nucleation sites likely contribute to the observed temperature differences.

The crystallization enthalpy for both the ERY-A and MAN-A formulations was determined from the thermographic data by integrating the difference between the measured temperature profile and a fitted cooling curve. This approach assumes that the heat capacity of the bulk binder is similar to that of the complete formulation, which is a strong simplification. Additionally, while the onset of crystallization is clearly visible in the temperature time data, the end of the event cannot be reliably distinguished from the natural cooling curve. For this reason, the endpoint of crystallization was assigned manually and conservatively. These values should be interpreted only as rough estimates due to the uncertain heat-capacity assumption and the manually assigned endpoint. They are included solely to provide qualitative insight into the extent of crystallization during cooling.

Both the uncertainty in defining the end of crystallization and the likely inaccurate heat capacity assumption led to approximate crystallization enthalpies of 40 ± 23 J/g for MAN-A and 58 ± 26 J/g for ERY-A. This is only a small fraction of the expected crystallization enthalpy when compared to the melting enthalpy of the formulations as determined by DSC (259.6 ± 2.3 J/g for MAN-A and 299.9 ± 2.1 J/g for ERY-A). The IR imaging reveals that a significant fraction of the formulations recrystallizes directly upon cooling within the build chamber, yet the uncertainty in integrating the IR curves renders inconclusive whether the materials has crystallized partially or fully at this moment. XRPD (Fig. 6D-F) and DSC analysis (Fig. 7A-C) of the tablets confirmed that ERY-A and MAN A tablets had completely crystallized by the time of analysis, which is in line with the crystallization behavior of erythritol and mannitol [18].

Fig. 6E shows that the MAN-A formulation consists mainly of β-mannitol, while the tablets contain mostly the α-polymorph of mannitol. It is well known that different polymorphs can arise from different preparation methods [61]. The spray drying process used to create the mannitol particles favors the formation of the β-form, while crystallization from the melt created by the laser seems to favor the formation of the α-form. Cornel et al. show that the α-form is slightly more soluble than the β-form over a temperature range of 5 to 25 °C [62]. A slight increase in solubility is advantageous for an immediate-

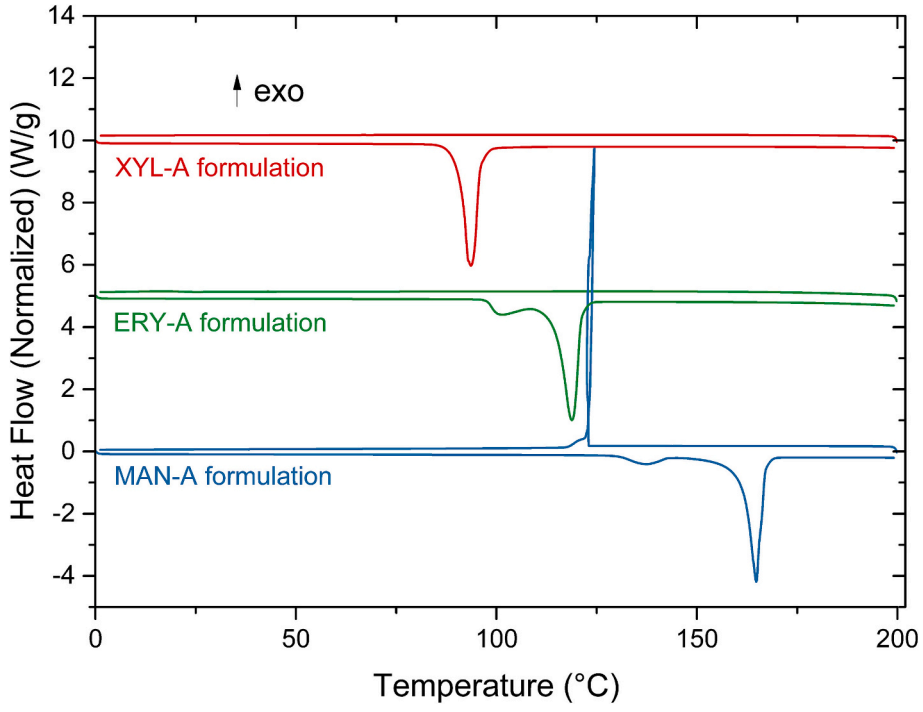


Fig. 5. DSC thermographs of the ERY-A, MAN-A, and XYL-A formulations, showing both heating and cooling curves. The looping observed in the mannitol cooling curve is caused by the strong exothermic crystallization of the mannitol, which temporarily offsets the programmed cooling rate of the DSC oven.

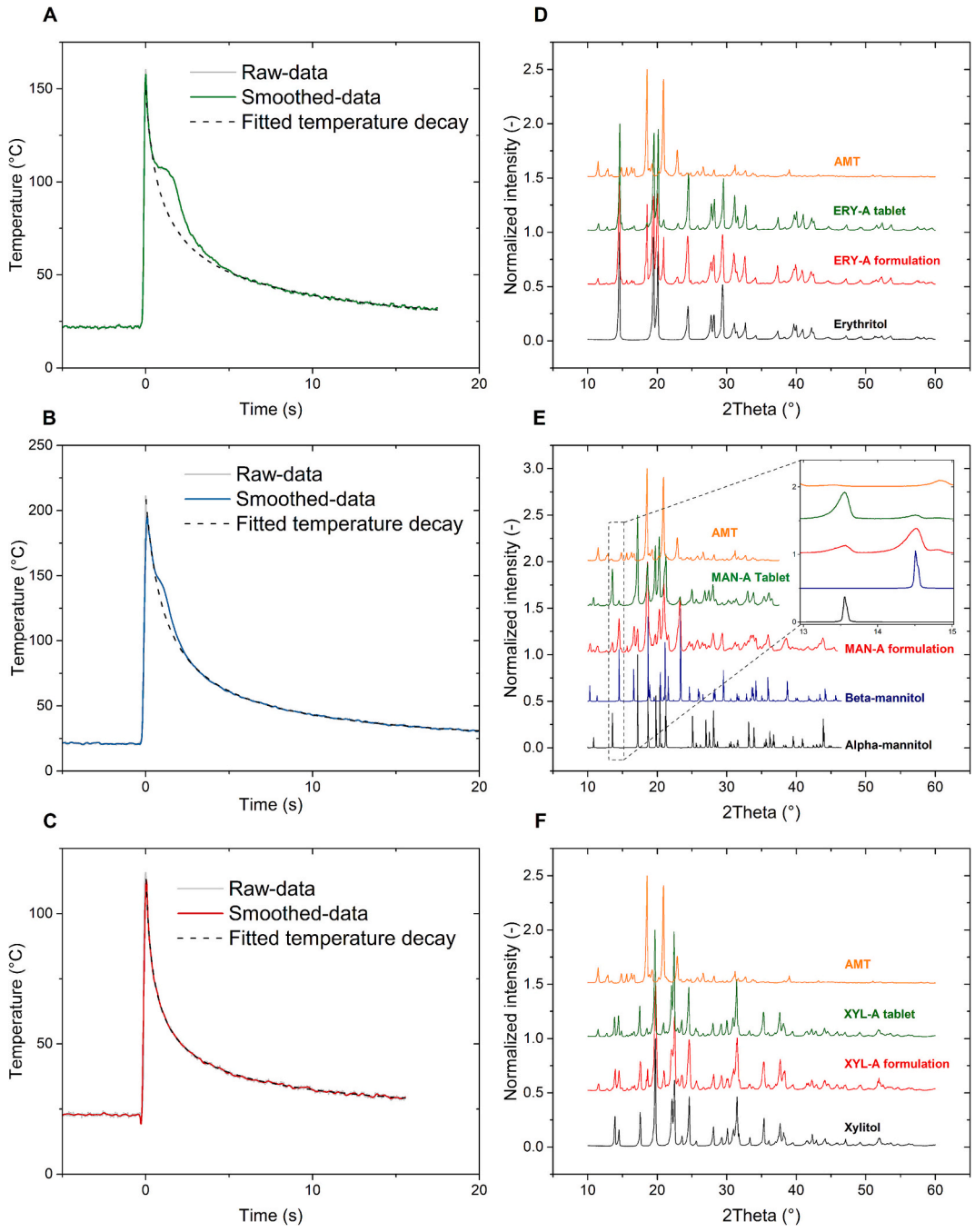


Fig. 6. Surface temperature profiles of the *E*-AMT (**A**), M-AMT (**B**), and X-AMT (**C**) formulations during manufacturing, extracted from a single pixel measurement. Panels D—F show measured X-ray powder diffractograms of: **(D)** xylitol, XYL-A formulation, XYL-A tablet, and AMT, **(E)** MAN-A formulation, MAN-A tablet, and AMT, together with predicted X-ray powder diffractograms of α- and ß-mannitol, **(F)** erythritol, ERY-A formulation, ERY-A tablet, and AMT [60].

release formulation, but the increase would be negligible.

The supercooling behavior of xylitol observed during the DSC measurement (Fig. 5) is corroborated by the thermographic data (Fig. 6C), which does not show a crystallization peak. As a result, the XYL-A tablets retained a semi-liquid core and bottom, which stuck to the build platform, preventing removal of intact tablets. To promote recrystallization, the scan speed was doubled for the first two layers, lowering energy density by half. Consequently, powder melting of the first tablet layer was incomplete, and solid particles could serve as nucleation points for subsequent molten layers to crystallize. A ten-minute delay before extracting the XYL-A tablets allowed them to solidify. The X-ray powder diffractograms and DSC confirm that the xylitol and AMT crystallize over time within the tablet (Fig. 6F).

3.3.2. *Thermal stability of AMT during PBF-LB.*

The peak temperatures reached during PBF-LB as measured with the IR camera (Fig. 6A-C) are listed in Table 6. The degradation temperatures of AMT, erythritol, mannitol, and xylitol were determined with TGA (**Fig. S4**) and are listed in Table 6. The XYL-A tablets were produced well below the degradation temperatures of both AMT and xylitol at all times, indicating no risk of the degradation of AMT, as is confirmed by UPLC and visual inspection. The ERY-A tablets were manufactured close to the degradation temperature of AMT, as indicated by the IR camera. However, UPLC content uniformity analysis confirmed that no detectable amounts of degradation products were formed.

For the MAN-A formulation, the AMT degradation temperature was exceeded by approximately 45 °C according to the IR data, suggesting

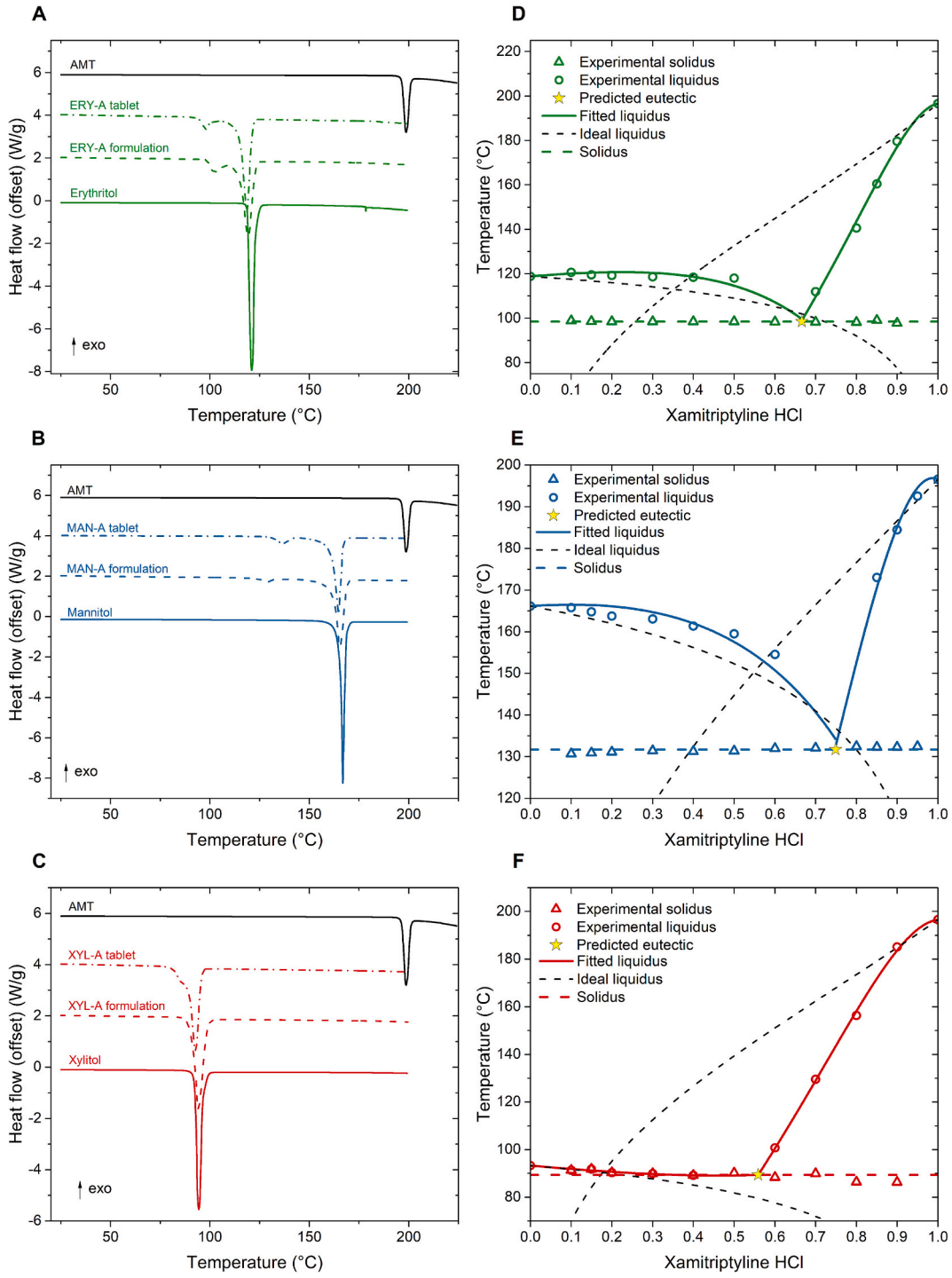


Fig. 7. DSC thermograms of the ERY-A (A), MAN-A (B), and XYL-A (C) compared to their individual components, showing only the heating curves. Panel (D—F) show binary phase diagrams of erythritol (D), mannitol (E) and xylitol (F) with AMT. The X-axis indicates the AMT weight fraction from 0 to 1, with the respective sugar alcohol decreasing from 1 to 0 accordingly. The star indicates the predicted eutectic point.

that some degree of degradation could occur. The slight discoloration of the MAN-A tablets supports this possibility. Nevertheless, the AMT content remained unchanged, and UPLC analysis did not reveal any additional peaks. Notably, mannitol itself is known to develop a yellow tint at temperatures of 180 °C and above [48], which may also contribute to the observed discoloration. Although the analytical method was not explicitly developed to be stability-indicating, the formation of substantial amounts of degradation products is unlikely. The material is exposed to temperatures above the AMT degradation point for only a brief period (~0.5 s), and the protective nitrogen atmosphere further reduces the potential for thermally induced oxidation.

Peak temperatures could be lowered by reducing the overall energy density used for manufacturing the MAN-A tablets. Peak temperatures can also be lowered by reducing both the laser power and scanning speed proportionally, which lowers the peak temperatures while maintaining the total energy density input, thus preserving particle fusion efficiency without promoting degradation. This strategy was not explored in this work [63].

Table 6
Degradation temperatures determined by TGA ($n = 3$) of the components of the formulations and the peak temperatures reached during the additive manufacturing process.

Formulation	Degradation temperatures via TGA (n = 3)		Peak temperature during the process ($n = 20$)
	AMT	**Sugar alcohol**	
ERY-A	162.1 ± 3.0 °C	152.9 ± 2.0 °C	151 ± 13 °C
MAN-A		226.1 ± 2.5 °C	207 ± 14 °C
XYL-A		192.2 ± 5.1 °C	111 ± 7 °C

3.3.3. Sugar alcohol–AMT eutectics

In Fig. 7A-C, it can be seen that during the heating cycle in the DSC, the AMT in the ERY-A, MAN-A, and XYL-A formulations and their corresponding tablets does not melt at its original melting temperature of 196.6 ± 0.4 °C. Instead, a eutectic mixture forms, where AMT melts at a lower temperature in a specific ratio with the sugar alcohol. Following this, the excess sugar alcohol that is still present beyond the eutectic composition melts at a temperature lower than its pure melting point, likely due to partial miscibility or continued interaction with the drug in the molten phase.

For each of the three sugar alcohol-AMT combinations, a binary phase diagram was constructed (Fig. 7D-F). The dashed black line in each panel represents the liquidus line assuming ideal behavior, calculated using Eq. 3 (SVL equation). The solid black line indicates the solidus, determined by averaging the experimentally measured solidus points, which are marked with triangles. To account for deviations from ideality, the experimental liquidus data were asymmetrically fitted using the SVL equation, incorporating activity coefficients as described in **section 2.4.4**. The drug-rich (left) and sugar alcohol–rich (right) branches of the liquidus lines were fitted independently, yielding two fitting parameters per side. Additionally, a fifth fitting parameter, common to both branches, was used to fit both the left and right branches of the fitted liquidus through a common point on the solidus, predicting the eutectic. The approximated mass ratios of the eutectics for each of the sugar alcohol – AMT combinations are listed in Table 7 with the original melting temperature of the components and the melting temperature of the eutectic mixture.

All three experimentally determined phase diagrams exhibit significant deviations from the ideal behavior predicted by Eq. 3, indicating the presence of strong interactions between AMT and the sugar alcohols. The sugar alcohols all display slightly positive deviations from ideal mixing behavior, whereas AMT shows strong negative deviations. Kollau et al. and others have partially explained such deviations through the presence and interactions of hydrogen bond donors and acceptors [64–66]. The deviations for these phase diagrams (Fig. 7D-F) can also be explained through hydrogen bonding. Sugar alcohols contain many hydrogen-bond donors and acceptors, allowing them to form extensive hydrogen-bond networks. In its crystal structure, the hydrochloride salt AMT exists as a protonated amine, with the nitrogen already involved in an N—H bond. This allows AMT to mix well with sugar alcohols, which can accept hydrogen bonds from the protonated AMT. However, from the sugar alcohols perspective, AMT is a weaker hydrogen-bonding partner than the polyols themselves, offering fewer interaction sites. This imbalance results in an unfavorable deviation from ideal mixing on the sugar alcohol's side, indicated by the positive deviation of the sugar alcohols and the negative deviation of the AMT in the phase diagram. The interaction of the polyols and AMT (a type 3 eutectic) has been described in more detail by Smith et al. [66]. The mannitol-AMT eutectic has been patented for pharmaceutical application, although it is unclear what the exact purpose of the patent is [67].

Table 7
Eutectic melting temperatures and compositions compared with the melting temperatures of the individual components for three binary systems of AMT with erythritol, mannitol and xylitol.

Binary mixture	Eutectic AMT mass ratio	$T_{m,\ binder}$ (°C, n = 3)	$T_{m,\ AMT}$ (°C, n = 3)	$T_{m,eutectic}$ (°C)
ERY-A	0.67	118.8 ± 0.03	196.6 ± 0.4	98.4
MAN-A	0.75	166.1 ± 0.8		131.7
XYL-A	0.56	93.3 ± 0.2		89.38

One advantageous change observed in eutectic mixtures is that the API can melt along with the binder material, allowing it to actively participate in forming a unified, solid tablet structure. By forming a eutectic, APIs with relatively high melting temperatures can contribute directly to tablet cohesion. From the mannitol-AMT phase diagram, it can be deduced that 75% AMT loading can be achieved while maintaining the reduced melting temperature of the AMT. A second advantage would be that, as the formulation approaches the eutectic ratio, its overall melting temperature decreases. This would enable processing at lower temperatures, thereby reducing thermal exposure of the API and sugar alcohol binders, as well as lowering energy consumption.

To test these potential advantages, a formulation containing AMT and mannitol at their eutectic ratio was prepared ((24.75% mannitol, 74.25% AMT, 1% colloidal silica) MAN-75 A), and 25 tablets (6.9 mm diameter) were manufactured at an energy density of 0.5 J/mm^3. The resulting tablets had an average weight of 89.3 ± 1.3 mg and a tablet tensile strength of 0.80 ± 0.08 MPa. For comparison, the standard 15 wt % AMT formulation (MAN-A) was manufactured under the same 0.5 J/mm^3 energy density, yielding tablets with an average weight of 76.5 ± 1.6 mg and a tensile strength of less than half of that (0.34 ± 0.04 MPa).

Fig. 8A shows the thermograms of MAN-A (15% AMT) and MAN-75 A (74.25% AMT), with the peak temperatures observed with IR imaging during PBF-LB at 0.5 J/mm^3 marked with an X. As can be observed, the peak temperature during manufacturing of MAN-75 A exceeds the melting point of the eutectic, meaning that the formulation completely melts as there are no components in excess of the eutectic. The peak temperature during manufacturing of MAN-A only exceeded the eutectic melting point, and not the melting point of the excess mannitol. The MAN-A formulation, therefore, only partially melts under these conditions. The incomplete melting and fusion of the MAN-A formulation explain the lower tensile strength and the lower tablet weights. Upon liquification, the AMT/mannitol powder is condensed and forms a small depressed area in the surrounding powder bed, which is filled with new powder at the next powder spreading cycle. Tablets with a higher molten fraction (MAN-75 A) will be heavier and stronger than ones in which powder is only partially molten (MAN-A) during the fusion process. These findings demonstrate that mixtures formulated at or near the eutectic composition require less energy to achieve complete melting, enabling manufacturing of strong and dense tablets at lower energy densities.

Additionally, the MAN-A and MAN75 A formulations were manufactured at an energy density of 1.0 J/mm^3. This is the same energy density at which the MAN-A tablets studied in **Section 3** were manufactured. From the IR imaging, it was determined that the measured peak temperatures reached roughly **267.6 ± 12.3 °C** for the MAN-75 A formulation and **227.7 ± 18.9 °C** for the MAN-A formulation. These temperatures are sufficient to melt both the eutectic fraction and the excess mannitol present in the formulations. However, when the AMT-rich MAN-75 A formulation is processed at this high energy density (1 J/mm^3), thermal degradation is observed, appearing as black spots on the manufactured tablets, while the MAN-A tablets show no such signs of degradation (Fig. 8B). The higher peak temperature of the MAN-75 A formulation compared to the MAN-A formulation under identical laser conditions can be attributed to differences in melting enthalpies of the components. AMT exhibits a lower melting enthalpy (100.5 J/g) than mannitol (282.5 J/g). Consequently, in AMT-rich formulations, less energy is required for melting, allowing a greater fraction of the absorbed laser energy to contribute to further heating of the material and subsequently degradation. In mannitol-rich formulations, a larger portion of the laser energy is consumed during melting, leaving less

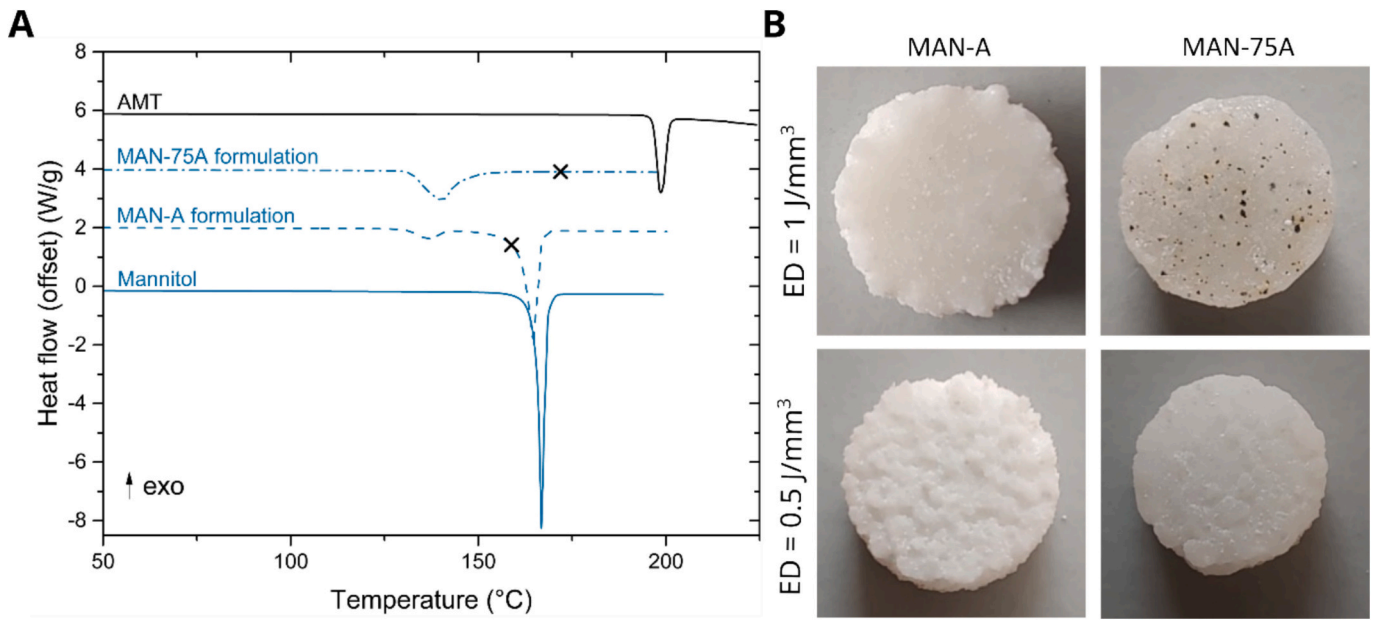


Fig. 8. (A) DSC thermograms of the MAN-A and MAN-75 A formulations. The two x's give a rough indication of the maximum temperature reached during manufacturing with an energy density of 0.5 J/mm^2. (B) Tablets manufactured from the MAN-A (25% AMT) MAN-75 A (75% AMT) formulation at an energy density of 0.5 J/mm^3 and 1.0 J/mm^3.

energy available to increase the temperature. Our findings hence also indicate the importance of carefully studying the thermal behavior of the materials during LBPF, since heat excess as well as incomplete melting can affect the final tablet in a negative way.

These findings confirm the above hypothesized advantages that formulations manufactured at or near their eutectic composition led to better fusion and mechanical strength at lower energy inputs. While the use of APIs as binders has been reported previously, the deliberate lowering of the API's melting point through eutectic formation to improve processability in PBF-LB appears to be a novel strategy.

4. Conclusion

For the first time, we demonstrate that PBF-LB of tablets can be achieved without preheating the powder bed. Moreover, we introduced a simple and robust dose calibration method for PBF-LB, enabling the manufacturing of any dose within the 5–45 mg range. Five different dose strengths were successfully manufactured with each sugar alcohol, showing low weight variation within and between batches. Tablets created with mannitol and erythritol had acceptable content uniformity across all doses, while xylitol-based tablets showed acceptable uniformity at all but the lowest (5 mg) dose, likely due to insufficient blend homogeneity. All tablets displayed high mechanical strength (tensile strength >1.4 MPa), rapid disintegration (< 6 min), and immediate AMT release (Q80 < 10 min).

A key mechanistic finding was the formation of eutectic mixtures between AMT and the sugar alcohols, which significantly lowered the melting point of AMT (by 67 °C with mannitol, by 98 °C with erythritol, and by 107 °C with xylitol). This enabled AMT to actively contribute to binding rather than behaving as an inert component. Additionally, it was shown that manufacturing a formulation at the eutectic point requires less energy due to the melting point depression.

While sorbitol proved to be unsuitable in preliminary studies for this work due to pronounced and prolonged supercooling, other sugar alcohols such as lactitol and maltitol warrant further investigation. A limitation of this work was the necessity to fuse the tablets to the build plate to avoid balling and warping. This requirement reduced the number of tablets that could be manufactured per batch, since stacking in the z-direction was not possible. Future research should focus on developing process strategies that circumvent balling, dragging, and warping, thereby improving scalability of PBF-LB tablet manufacturing.

Overall, this work establishes sugar alcohols as effective alternatives to polymeric binders in PBF-LB. Their combination of strong binding, rapid dissolution, and favorable eutectic interactions expands the excipient toolbox, opening new opportunities for flexible and efficient formulation development.

CRediT authorship contribution statement

Wessel Kooijman: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Florian Engelsing:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Robbert Jan Kok:** Writing – review & editing, Supervision. **Julian Quodbach:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors have no conflict of interest to declare.

Acknowledgements

The authors are grateful for the materials supplied by Roquette and Evonik.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jmapro.2026.02.076.

Data availability

All data used in this study are openly available at https://public.yoda.uu.nl/science/UU01/JGGXL1.html.

References

[1] Castro-Balado A, Varela-Rey I, Mejuto B, Mondelo-García C, Zarra-Ferro I, Rodríguez-Jato T, et al. Updated antimicrobial dosing recommendations for obese patients. Antimicrob Agents Chemother 2024;68. https://doi.org/10.1128/aac.01719-23.

[2] Niederberger E, Parnham MJ. The Impact of Diet and Exercise on Drug Responses. Int J Mol Sci 2021;22:7692. https://doi.org/10.3390/ijms22147692.

[3] Micaglio E, Locati ET, Monasky MM, Romani F, Heilbron F, Pappone C. Role of Pharmacogenetics in Adverse Drug Reactions: An Update towards Personalized Medicine. Front Pharmacol 2021;12. https://doi.org/10.3389/fphar.2021.651720.

[4] Zevin S, Benowitz NL. Drug Interactions with Tobacco Smoking. Clin Pharmacokinet 1999;36:425–38. https://doi.org/10.2165/00003088-199936060-00004.

[5] Breitkreutz J, Boos J. Paediatric and geriatric drug delivery. Expert Opin Drug Deliv 2007;4:37–45. https://doi.org/10.1517/17425247.4.1.37.

[6] Soldin OP, Mattison DR. Sex Differences in Pharmacokinetics and Pharmacodynamics. Clin Pharmacokinet 2009;48:143–57. https://doi.org/10.2165/00003088-200948030-00001.

[7] Swen JJ, van der Wouden CH, Manson LE, Abdullah-Koolmees H, Blagec K, Blagus T, et al. A 12-gene pharmacogenetic panel to prevent adverse drug reactions: an open-label, multicentre, controlled, cluster-randomised crossover implementation study. Lancet 2023;401:347–56. https://doi.org/10.1016/S0140-6736(22)01841-4.

[8] Trivedi M, Jee J, Silva S, Blomgren C, Pontinha VM, Dixon DL, et al. Additive manufacturing of pharmaceuticals for precision medicine applications: A review of the promises and perils in implementation. Addit Manuf 2018;23:319–28. https://doi.org/10.1016/j.addma.2018.07.004.

[9] Kopp S-P, Medvedev V, Tangermann-Gerk K, Wöltinger N, Rothfelder R, Graßl F, et al. Electrophotographic 3D printing of pharmaceutical films. Addit Manuf 2023;73:103707. https://doi.org/10.1016/j.addma.2023.103707.

[10] Jørgensen AK, Oh YC, Li H, Treffer D, Parhizkar M, Goyanes A, et al. Expediting 3D printed medication development using vacuum compression moulding. J Control Release 2025;383:113766. https://doi.org/10.1016/j.jconrel.2025.113766.

[11] Wang N, Shi H, Yang S. 3D printed oral solid dosage form: Modified release and improved solubility. J Control Release 2022;351:407–31. https://doi.org/10.1016/j.jconrel.2022.09.023.

[12] Gueche YA, Sanchez-Ballester NM, Cailleaux S, Bataille B, Soulairol I. Selective Laser Sintering (SLS), a New Chapter in the Production of Solid Oral Forms (SOFs) by 3D Printing. Pharmaceutics 2021;13. https://doi.org/10.3390/pharmaceutics13081212.

[13] Tabriz A, Junqueira LA, Kolipaka SS, Hui H-W, Boersen N, Roberts S, et al. Printabily of Pharmaceutical-Grade Polymers using Selective Laser Sintering with a CO2 Laser. AAPS PharmSciTech 2025;26:144. https://doi.org/10.1208/s12249-025-03141-4.

[14] Hess F, Kipping T, Weitschies W, Krause J. Understanding the interaction of thermal, rheological, and mechanical parameters critical for the processability of polyvinyl alcohol-based systems during hot melt extrusion. Pharmaceutics 2024;16:472. https://doi.org/10.3390/pharmaceutics16040472.

[15] Parikh T, Gupta SS, Meena A, Serajuddin A. Investigation of thermal and viscoelastic properties of polymers relevant to hot melt extrusion - III: Polymethacrylates and polymethacrylic acid based polymers. J Excip Food Chem 2014;5:56–64.

[16] Yang F, Su Y, Zhang J, DiNunzio J, Leone A, Huang C, et al. Rheology guided rational selection of processing temperature to prepare Copovidone–Nifedipine amorphous solid dispersions via hot melt extrusion (HME). Mol Pharm 2016;13:3494–505. https://doi.org/10.1021/acs.molpharmaceut.6b00516.

[17] Kooijman W, Levine VR, Kok RJ, Lindh J, Quodbach J. The effects of recycling pharmaceutical formulations in laser powder bed fusion 3D printing - the influence of physical phenomena on printing performance. Int J Pharm X 2025;10:100383. https://doi.org/10.1016/j.ijpx.2025.100383.

[18] Shao XF, Wang C, Yang YJ, Feng B, Zhu ZQ, Wang WJ, et al. Screening of sugar alcohols and their binary eutectic mixtures as phase change materials for low-to-medium temperature latent heat storage. (I): Non-isothermal melting and crystallization behaviors. Energy 2018;160:1078–90. https://doi.org/10.1016/J.ENERGY.2018.07.081.

[19] Shao X, Yang S, Chen C, Fan L, Yu Z. Temperature-Dependent Rheological Behaviors of Binary Eutectic Mixtures of Sugar Alcohols for Latent Heat Storage: A Comparative Study with Pure Sugar Alcohols. J Therm Sci 2021;30:2002–14. https://doi.org/10.1007/s11630-021-1279-9.

[20] Allahham N, Fina F, Marcuta C, Kraschew L, Mohr W, Gaisford S, et al. Selective Laser Sintering 3D Printing of Orally Disintegrating Printlets Containing Ondansetron. Pharmaceutics 2020;12:110. https://doi.org/10.3390/pharmaceutics12020110.

[21] Stanojević G, Adamov I, Mugoša S, Vukićević V, Ibrić S. Selective Laser Sintering of Atomoxetine Tablets: An Innovative Approach for Small-Scale. Personalized Production, Pharmaceutics 2025;17:794. https://doi.org/10.3390/pharmaceutics17060794.

[22] Gotoh A, Yamazoe E, Ito T, Koide Y, Yamada M, Shimada Y, et al. Effects of excipient properties on pharmaceutical printlet fabrication via selective laser sintering 3D printing. Adv Powder Technol 2024;35:104678. https://doi.org/10.1016/j.apt.2024.104678.

[23] Horváth ZM, Kukuls K, Frolova AJ, Žogota M, Buczkowska EM, Pētersone L, et al. Effervescent Tablet Preparation by Twin-Screw Melt Granulation with Sorbitol as a Melt Binder. Pharmaceutics 2025;17. https://doi.org/10.3390/pharmaceutics17050676.

[24] Lin L, Lin X, Shen L, Hong Y, Zhao L. Mannitol in direct compression: Production, functionality, critical material attributes and co-processed excipients. Int J Pharm 2025;676:125595. https://doi.org/10.1016/J.IJPHARM.2025.125595.

[25] Janssen PHM, Fathollahi S, Bekaert B, Vanderroost D, Roelofs T, Vanhoorne V, et al. Impact of material properties and process parameters on tablet quality in a continuous direct compression line. Powder Technol 2023;424:118520. https://doi.org/10.1016/J.POWTEC.2023.118520.

[26] EUROPEAN PARLIAMENT AND COUNCIL DIRECTIVE 94/35/EC of 30 June 1994 on sweeteners for use in foodstuffs. In: Official Journal of the European Communities; 1994.

[27] Younes M, Aquilina G, Castle L, Degen G, Engel K, Fowler PJ, et al. Re-evaluation of erythritol (E 968) as a food additive. EFSA J 2023;21. https://doi.org/10.2903/j.efsa.2023.8430.

[28] EFSA. Sweeteners. https://www.efsa.europa.eu/en/topics/topic/sweeteners. [Accessed 25 August 2025].

[29] Tennant DR. Potential intakes of total polyols based on UK usage survey data. Food Additives & Contaminants: Part A 2014;31:574–86. https://doi.org/10.1080/19440049.2014.886132.

[30] Grembecka M. Sugar alcohols—their role in the modern world of sweeteners: a review. Eur Food Res Technol 2015;241:1–14. https://doi.org/10.1007/s00217-015-2437-7.

[31] Witkowski M, Nemet I, Alamri H, Wilcox J, Gupta N, Nimer N, et al. The artificial sweetener erythritol and cardiovascular event risk. Nat Med 2023;29:710–8. https://doi.org/10.1038/s41591-023-02223-9.

[32] Witkowski M, Nemet I, Li XS, Wilcox J, Ferrell M, Alamri H, et al. Xylitol is prothrombotic and associated with cardiovascular risk. Eur Heart J 2024;45:2439–52. https://doi.org/10.1093/eurheartj/ehae244.

[33] Dean L, Kane M. Amitriptyline Therapy and CYP2D6 and CYP2C19 Genotype. 2012.

[34] College ter Beoordeling van Geneesmiddelen - Medicines Evaluation Board, AMITRIPTYLINEHYDROCHLORIDE, (n.d.). https://www.geneesmiddeleninformatiebank.nl/ (accessed October 13, 2025).

[35] Zorginstituut Nederland, amitriptyline, (n.d.). https://www.farmacotherapeutischkompas.nl/bladeren/preparaatteksten/a/amitriptyline (accessed October 13, 2025).

[36] van Os J, Groot PC. Outcomes of hyperbolic tapering of antidepressants. Ther Adv Psychopharmacol 2023;13. https://doi.org/10.1177/20451253231171518.

[37] Groot PC, van Os J. Successful use of tapering strips for hyperbolic reduction of antidepressant dose: a cohort study. Ther Adv Psychopharmacol 2021;11. https://doi.org/10.1177/20451253211039327.

[38] Moraczewski J, Awosika AO, Aedma KK. Tricyclic Antidepressants. 2024.

[39] Schröder Iw. Über die Abhängigkeit der Löslichkeit eines festen Körpers von seiner Schmelztemperatur. Z Phys Chem 1893;11U:449–65. https://doi.org/10.1515/zpch-1893-1134.

[40] van Laar J. Die Schmelz- oder Erstarrungskurven bei binären Systemen, wenn die feste Phase ein Gemisch (amorphe feste Lösung oder Mischkristalle) der beiden Komponenten ist. Z Phys Chem 1908;64U:257–97. https://doi.org/10.1515/zpch-1908-6417.

[41] Fell JT, Newton JM. Determination of Tablet Strength by the Diametral-Compression Test. J Pharm Sci 1970;59:688–91. https://doi.org/10.1002/JPS.2600590523.

[42] Spaght ME, Thomas SB, Parks GS. Some Heat-Capacity Data on Organic Compounds obtained with a Radiation Calorimeter. J Phys Chem 1932;36:882–8. https://doi.org/10.1021/j150333a009.

[43] Snow Z, Martukanitz R, Joshi S. On the development of powder spreadability metrics and feedstock requirements for powder bed fusion additive manufacturing. Addit Manuf 2019;28:78–86. https://doi.org/10.1016/j.addma.2019.04.017.

[44] Elbadawi M, Li H, Ghosh P, Alkahtani ME, Lu B, Basit AW, et al. Cold Laser Sintering of Medicines: Toward Carbon Neutral Pharmaceutical Printing. ACS Sustain Chem Eng 2024;12:11155–66. https://doi.org/10.1021/acssuschemeng.4c01439.

[45] Guo S, Li J, Zhang H, Zou L, Zhao W. Balling behavior of phenolic resin during selective laser sintering process. Mater Lett 2024;365:136447. https://doi.org/10.1016/j.matlet.2024.136447.

[46] Zheng Z, Krueger L, Harrop ACF, Ross BP, Popat A, Miles JA. 3D printed tablets for personalised dose titration of prednisone using selective laser sintering. Int J Pharm 2025;678:125698. https://doi.org/10.1016/j.ijpharm.2025.125698.

[47] Krueger L, Cao Y, Zheng Z, Ward J, Miles JA, Popat A. 3D printing tablets for high-precision dose titration of caffeine. Int J Pharm 2023;642:123132. https://doi.org/10.1016/j.ijpharm.2023.123132.

[48] Rodríguez-García M-M, Bayón R, Rojas E. Stability of D-mannitol upon Melting/Freezing Cycles under Controlled Inert Atmosphere. Energy Procedia 2016;91:218–25. https://doi.org/10.1016/j.egypro.2016.06.207.

[49] Tikhomirov E, Åhlén M, Di Gallo N, Strømme M, Kipping T, Quodbach J, et al. Selective laser sintering additive manufacturing of dosage forms: Effect of powder formulation and process parameters on the physical properties of printed tablets. Int J Pharm 2023;635:122780. https://doi.org/10.1016/j.ijpharm.2023.122780.

[50] Brennan MC, Keist JS, Palmer TA. Defects in Metal Additive Manufacturing Processes. J Mater Eng Perform 2021;30:4808–18. https://doi.org/10.1007/s11665-021-05919-6.

[51] Pitt KG, Heasley MG. Determination of the tensile strength of elongated tablets. Powder Technol 2013;238:169–75. https://doi.org/10.1016/j.powtec.2011.12.060.

[52] Lura A, Tardy G, Kleinebudde P, Breitkreutz J. Tableting of mini-tablets in comparison with conventionally sized tablets: A comparison of tableting properties

and tablet dimensions. Int J Pharm X 2020;2. https://doi.org/10.1016/j.ijpx.2020.100061.
[53] Poka MS, Milne M, Wessels A, Aucamp M. Sugars and Polyols of Natural Origin as Carriers for Solubility and Dissolution Enhancement. Pharmaceutics 2023;15:2557. https://doi.org/10.3390/pharmaceutics15112557.
[54] Rowe RC, Sheskey PJ, Owen SC. Handbook of pharmaceutical excipients. Pharmaceutical Press ; American Pharmacists Association; 2006.
[55] Narala S, Komanduri N, Nyavanandi D, Youssef AAA, Mandati P, Alzahrani A, et al. Hard gelatin capsules containing hot melt extruded solid crystal suspension of carbamazepine for improving dissolution: Preparation and in vitro evaluation. J Drug Delivery Sci Technol 2023;82:104384. https://doi.org/10.1016/j.jddst.2023.104384.
[56] Pawar JN, Fule RA, Maniruzzaman M, Amin PD. Solid crystal suspension of Efavirenz using hot melt extrusion: Exploring the role of crystalline polyols in improving solubility and dissolution rate. Mater Sci Eng C 2017;78:1023–34. https://doi.org/10.1016/j.msec.2017.04.055.
[57] Liu H, Jing J, Liu J, Wang X. Sugar alcohol-based phase change materials for thermal energy storage: Optimization design and applications. Renew Sust Energ Rev 2024;199:114528. https://doi.org/10.1016/j.rser.2024.114528.
[58] Liu C, Cheng Q, Li B, Liu X, Rao Z. Recent advances of sugar alcohols phase change materials for thermal energy storage. Renew Sust Energ Rev 2023;188:113805. https://doi.org/10.1016/j.rser.2023.113805.
[59] Yang S, Shi H-Y, Liu J, Lai Y-Y, Bayer Ö, Fan L-W. Supercooled erythritol for high-performance seasonal thermal energy storage. Nat Commun 2024;15:4948. https://doi.org/10.1038/s41467-024-49333-7.
[60] Fronczek FR, Kamel HN, Slattery M. Three polymorphs (α, β, and δ) of D -mannitol at 100 K. Acta Crystallogr C 2003;59:o567–70. https://doi.org/10.1107/S0108270103018961.
[61] Yang Y, Liu J, Hu A, Nie T, Cheng Z, Liu W. A Critical Review on Engineering of d-Mannitol Crystals: Properties, Applications, and Polymorphic Control. Crystals (Basel) 2022;12:1080. https://doi.org/10.3390/cryst12081080.
[62] Cornel J, Kidambi P, Mazzotti M. Precipitation and Transformation of the Three Polymorphs of D - Mannitol. Ind Eng Chem Res 2010;49:5854–62. https://doi.org/10.1021/ie9019616.
[63] Ren Q, Chen C, Lu Z, Wang X, Lu H, Yin S, et al. Effect of a constant laser energy density on the evolution of microstructure and mechanical properties of NiTi shape memory alloy fabricated by laser powder bed fusion. Opt Laser Technol 2022;152:108182. https://doi.org/10.1016/J.OPTLASTEC.2022.108182.
[64] Kollau LJBM, Vis M, van den Bruinhorst A, Esteves ACC, Tuinier R. Quantification of the liquid window of deep eutectic solvents. Chem Commun 2018;54:13351–4. https://doi.org/10.1039/C8CC05815F.
[65] Abbott AP, Capper Glen, Davies DL, Rasheed RK, Tambyrajah Vasuki. Novel solvent properties of choline chloride/urea mixturesElectronic supplementary information (ESI). available: spectroscopic data. See, http://www.rsc.org/suppdata/cc/b2/b210714g/.
[66] Smith EL, Abbott AP, Ryder KS. Deep Eutectic Solvents (DESs) and Their Applications. Chem Rev 2014;114:11060–82. https://doi.org/10.1021/cr300162p.
[67] Colombo P, Nebuloni M. Eutectic formulations of amitriptyline hydrochloride and mannitol. 2014.